

Motivation of Project

The objective is to build a systematic research pipeline for cross-sectional FX strategies, with an emphasis on how signals and portfolio construction behave under realistic implementation constraints.

The framework is grounded in two ideas: cross-sectional factor premia in the sense of Stephen Ross, and state-dependent variation in expected returns as emphasized by John H. Cochrane. The pipeline is designed to test both clean factor exposures and conditional adjustments within a single, consistent structure.

The focus is on constructing a modular, fully causal workflow where each stage from data through portfolio returns is explicitly defined and reproducible, enabling systematic and credible evaluation of signals, conditioning logic, and portfolio transformations.

The end goal is to obtain feedback on how this design compares to systematic strategies implemented in practice, with priority placed on clarity, repeatability, and preserving out-of-sample validity rather than maximizing out-of-sample performance.

High-Level Overview

The pipeline builds a systematic FX research workflow from explicitly saved parquet deliverables. FX prices are downloaded from Yahoo Finance, converted into a consistent USD-per-foreign-currency convention using fixed inversion rules, and combined with FRED rate data. Rate data is shifted by one business day and then forward-filled before being used. Returns are computed as log spot returns plus the lagged interest-rate carry component, then aligned to the system dataframe and saved as `fx_system_df_lagged.parquet` and `fx_returns_df_lagged.parquet`.

The signal layer produces three standalone FX signals from the lagged system dataframe. Carry is the cross-sectional rate differential between each foreign currency and USD. Trend is a 128-day log price change. Value is implemented as the negative difference between a 50-day and 200-day moving average of log price, so it is a price-based reversal proxy rather than a PPP valuation signal. Each signal is aligned to returns, filtered for minimum cross-sectional coverage, cross-sectionally normalized, evaluated with 21-day forward return ICs, and saved as a separate parquet file.

Feature discovery is implemented as a separate macro and market-state research layer. It loads the lagged FX deliverables, then builds equity, commodity, volatility, FX volatility, and funding/liquidity feature panels. Features are generated from rolling transformations such as trends, changes, z-scores, change z-scores, persistence, breadth, dispersion, volatility regimes, funding spreads, liquidity measures, and curve slopes. Feature testing uses future FX returns only for diagnostics. Production state construction is walk-forward: filtered feature panels are reduced into one PCA state per category using only trailing 252-day windows, then saved as `states_df.parquet`.

Portfolio construction loads the saved carry, trend, and value signals into a MultiIndex signal panel. Standalone sleeves are built by cross-sectionally demeaning each signal, gross-normalizing weights, applying a one-day execution lag, and subtracting transaction costs based on 0.5 times absolute weight changes. Combined portfolios are then tested using equal signal weights, inverse-volatility sleeve allocation, regime scalar-weighted sleeve allocation, and an inverse-volatility plus scalar overlay. Sleeve scalars are estimated from expanding historical relationships between state variables and sleeve PnL, smoothed, clipped, and shifted by one day before use. Final portfolio weights are gross-normalized, lagged before return application, charged transaction costs, and saved with diagnostics.

Backtesting is run through realized portfolio PnL. PnL uses shifted weights against aligned FX returns. Net returns deduct turnover costs. Diagnostics report gross and net Sharpe, volatility, drawdown, skew, total return, and annualized turnover. The diagnostics notebook reads the saved portfolio outputs and analyzes cumulative PnL, drawdowns, rolling Sharpe, turnover, weight behavior, sleeve allocations, and return attribution from the stored parquet files.

Design Principles

Causality

The core pipeline is designed around explicit as-of handling. FRED rates are shifted before forward-filling, the main data files are labeled and saved as lagged deliverables, and all portfolio PnL calculations use `weights.shift(1)` before applying returns. Walk-forward PCA states use trailing windows only, and sleeve scalars are shifted one day before portfolio use. Forward returns are used in signal and feature notebooks for IC evaluation, but those forward returns are diagnostic outputs, not inputs to the traded portfolio weights.

Robustness

Robustness comes from simple economically grounded signal definitions, repeated alignment checks, minimum cross-sectional coverage filters, cross-sectional normalization, gross normalization, transaction cost deductions, and turnover measurement. The portfolio layer does not rely on unconstrained optimized weights. It uses demeaning for dollar neutrality, gross exposure normalization, lagged inverse-volatility sleeve allocation, clipped state-based scalars, and final gross normalization.

Modularity

The pipeline is modular because each stage writes concrete artifacts that the next stage loads directly. Data construction saves lagged FX system and returns files. Signal notebooks save carry, trend, and value signal files. Feature discovery saves cleaned feature panels and walk-forward state files. Portfolio construction loads those signal and state files, builds portfolio weights and PnL, then saves diagnostics artifacts for the diagnostics notebook. This makes the pipeline fixed and reproducible while allowing individual stages to be changed without rewriting the entire workflow.

Results of Signal Construction (Carry, Trend and Value)

IC Mean	IC Median	IC > 0 %	IC IR	IC Mean	IC Median	IC > 0 %	IC IR	IC Mean	IC Median	IC IR
0.04987	0.071429	0.517564	0.099144	0.003758	0.0	0.485474	-0.006955	0.035695	0.071429	0.063903

Raw Trend construction is very weak. I need something better than cross sectional ranking based on log 128 day change.

Passing Features that get compressed into states via PCA due to passing IC statistics with FX returns

Equity feature count: 10

Equity features: ['EZU_z_126', 'ENZL_z_63', 'EWC_z_126', 'ENZL_z_126', 'EWA_z_126', 'EWJ_z_126', 'EZU_z_63', 'EWJ_z_63', 'EWA_z_63', 'EWL_z_63']

Equity features chosen show that the signal is consistently picking up medium to long horizon relative strength across developed international equities, with a clear bias toward persistence Structurally, this

implies the equity sleeve is acting as a slow moving macro proxy, capturing broad risk appetite and regional divergence, which aligns well with FX where these same dynamics often transmit through capital flows and currency strength.

Commodity feature count: 10

Commodity features: ['DBC_z_126', 'DBC_z_63', 'GLD_z_126', 'GLD_z_63', 'DBC_z_21', 'GLD_z_21', 'DBC_persist_21', 'USO_persist_21', 'GLD_trend_126', 'DBC_trend_63']

Commodity features chosen show that the signal is consistently picking up medium to long horizon trend and persistence across broad commodity indices and gold, with a secondary sensitivity to shorter term continuation Structurally, this implies the commodity sleeve is acting as a real asset and inflation proxy, capturing directional moves in global growth and pricing pressures, which aligns with FX through terms of trade shifts and commodity currency dynamics

Vol feature count: 9

Vol features: ['VIX_vol_of_vol_21', 'FX_VOL_z_63', 'VIX_vol_of_vol_z', 'VIX_z_63', 'VIX_low', 'VIX_high', 'VIX_spike', 'VIX_crush', 'FX_vol_of_vol_21']

Vol features chosen show that the signal is consistently picking up regime shifts and convexity in volatility rather than directional trends, with emphasis on volatility of volatility and discrete state transitions Structurally, this implies the vol sleeve is acting as a market stress and regime detection layer, capturing transitions between risk on and risk off environments, which aligns with FX through changes in carry attractiveness, de risk flows, and cross asset correlation breakdowns

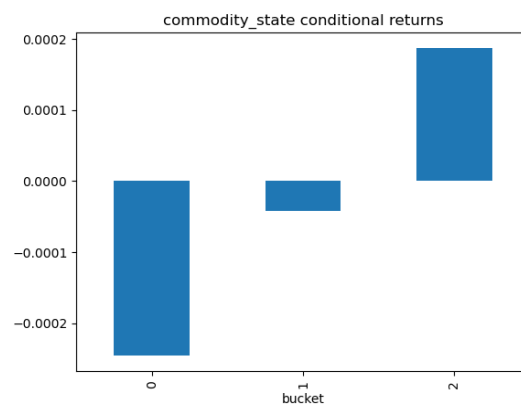
Funding feature count: 10

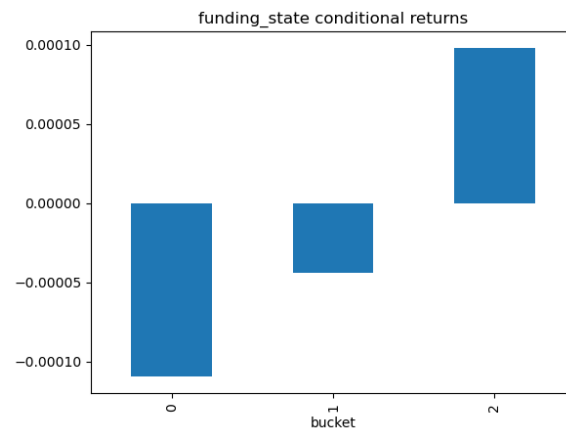
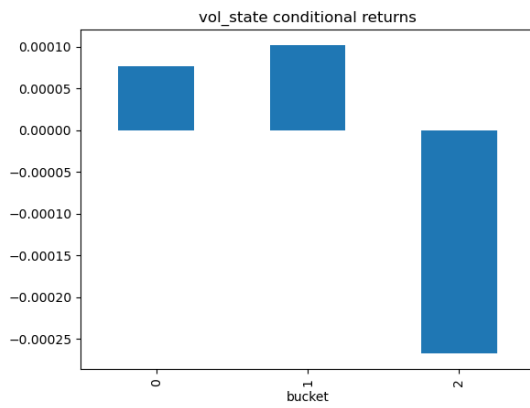
Funding features: ['EFFR_z_63', 'STLFSI_z_63', 'EFFR_chg_z_21', 'UST_3M_spike', 'UST_10Y_spike', 'slope_10_2_spike', 'UST_2Y_crash', 'slope_10_3m_spike', 'UST_2Y_spike', 'slope_10_3m_crash']

Funding features chosen show that the signal is consistently picking up shifts in short rate levels, liquidity stress, and yield curve dynamics, with a focus on both gradual changes and abrupt regime events Structurally, this implies the funding sleeve is acting as a policy and liquidity proxy, capturing the cost and availability of capital, which aligns with FX through interest rate differentials, carry conditions, and global funding stress transmission

I plan to enhance this section of my pipeline the most when I get better data quality.

Features vs Returns (monotonic behavior is desirable)





Equity and commodity good. Funding a bit weaker. Vol, not very good.

Portfolio Construction Results

```
✓ # EQUAL WEIGHT COMBINED SIGNAL PORTFOLIO ...
```

	GrossSharpe	NetSharpe	Turnover
equal_signal_combo	0.365227	0.238197	14.830101


```
✓ # INVERSE VOL SLEEVE COMBO PORTFOLIO ...
```

	GrossSharpe	NetSharpe	Turnover
inverse_vol_sleeve_combo	0.435324	0.327943	11.89616


```
✓ # SCALAR WEIGHTED SLEEVE COMBO PORTFOLIO ...
```

	GrossSharpe	NetSharpe	Turnover
scalar_weighted_sleeve_combo	0.307787	0.185696	13.56133


```
✓ # INVERSE VOL × SCALAR OVERLAY SLEEVE COMBO PORTFOLIO ...
```

	GrossSharpe	NetSharpe	Turnover
inverse_vol_scalar_overlay_combo	0.395305	0.285618	12.151762

Stand alone inverse volatility is currently the best performing portfolio construction method, delivering a 0.32 net Sharpe.

The scalar tilts and inverse volatility with scalar tilts underperforms this baseline, indicating that my scalar layer is not yet additive and is instead diluting a robust risk allocation framework of raw signal.

NEXT STEPS

Fixing Trend (doesn't require more data)

Trend is currently the weakest core signal and needs to be rebuilt.

Signal Construction

Evaluate whether directly interacting signals with features produces more stable and predictive structure relative to the current regime scalar approach. This involves generating interaction variants, reducing redundancy where necessary, and testing them against the existing framework on equal footing to determine whether conditioning should live inside the signal rather than as an overlay.

Data Expansion

Expand beyond current Yahoo Finance and FRED constraints to improve data quality and coverage, with a focus on incorporating institutional-grade sources for rates and funding data. Develop a true FX value factor using PPP or related macro inputs. Extending the universe to include emerging market FX would increase cross-sectional breadth and improve opportunity for relative value and carry dispersion.

Live Monitoring

Develop a framework for tracking signal decay and stability in a live setting, including ongoing evaluation of IC, dispersion, and performance consistency to ensure signals remain valid outside of the research environment.